The pair singular series of a polynomial IV: the second spectrum

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Abstract

Part III reduced the Cesàro form of the off-diagonal of the pair singular series of an irreducible quadratic f to "pieces" $\mathcal{P}_u(Y)$, one for each way the two roots of f can agree on a part u of the modulus, and showed that everything except the pieces' "windows" is understood. This paper is about what the pieces actually are. Numerically each piece is $O(\sqrt{Y})$, and $\mathcal{P}_u(Y)/\sqrt{Y}$ oscillates in $\log Y$ at the Laplace eigenvalues of the *even* Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$, with amplitudes governed by the values of the forms at the Heegner points, or their integrals over the closed geodesics, of discriminant D: a second spectrum, next to the zeros of ζ_K that govern the diagonal (part II). We explain this by an explicit formula: the piece is a twisted Walfisz sum whose Dirichlet series is a sum over frequencies of Dirichlet series of Salié sums, which the Kuznetsov formula of half-integral weight continues to the half-line with poles at $\frac{1}{2} \pm it_j$, and the Katok–Sarnak correspondence identifies the residues as periods of Maass forms. The same law, and the same parameter-free prediction of its phases, holds for a barer and more classical object: the partial sums of the Weyl sums of the congruence $b^2 \equiv D$, which Hooley bounded by $X^{3/4+\varepsilon}$ and which we find to exhibit square-root cancellation with an explicit spectral expansion. We prove it for $u = 1$: the root pairs are Heegner points, the piece is a sum of one test function over a modular orbit, the convergence obstruction and the passage to L^2 are resolved by a truncation and by the symmetry of the orbit, the constant and continuous contributions vanish or are negligible because the seed has vanishing horizontal mean, the arising Bessel integral is evaluated exactly by a Mellin–Barnes contour shift, and the spectral sum converges absolutely for Riesz means of order at least two; discuss the uniformity in u that the Cesàro conjecture of part I would need, and report the numerical evidence, including phase predictions with no free parameter that agree to a median of 0.012 radians.

1 Introduction

In plain terms. Parts I and II showed that the main part of the fluctuation of prime values of a polynomial is governed by the zeros of a Dedekind zeta function. The remaining part, the "off-diagonal" of part III, turns out to be governed by a second, entirely different spectrum: the Laplace eigenvalues of the modular surface, the same numbers that govern the distribution of the roots of $x^2 \equiv D \pmod{d}$ (Hooley, Bykovskiĭ, Duke–Friedlander–Iwaniec). Only half of them appear, those of the even forms, and the size of each contribution is the value of the corresponding eigenfunction at a special point, or its integral along a special closed curve, determined by the discriminant of the polynomial. We found this in the data before we had the theory; the theory, once written down, predicts exactly the pattern of presence and absence that the data show.

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Set-up. Notation as in part III: $f = t^2 + bt + c$, $D = b^2 - 4c$, $\lambda(p) = p/(p - 2\omega(p))$; for squarefree u with $\omega(u) \geq 1$ and $Q_u(x) = u^2x^2 - D$,

$$\mathcal{P}_u(Y) = \sum_{h\leq Y}(Y-h)\big(F_u(Q_u(h))-\mathbb{E}F_u\big)-\tfrac{1}{2}(\mathbb{E}F_u-1)Y, \qquad F_u(n) = \sum_{\substack{d\mid n,\ d>1\\ d\text{ admissible}}} \frac{\lambda(d)}{d}, \qquad \mathbb{E}F_u = \sum_{\substack{d>1\\ d\text{ adm}}} \frac{\lambda(d)\omega(d)}{d^2},$$

admissible meaning squarefree, all primes split, coprime to $2Du$. Part III, Theorem A$'$: $\mathrm{Off}^*_f(H) = c_{\mathrm{off}}(f)H + \sum_{u\leq H^{2/3+\varepsilon}} w(u)\mathcal{W}_u(H/u; \log H) + O(H(\log H)^{1-c})$, and the Cesàro form of Conjecture 1 of part I is equivalent to the windows being $o(H \log H)$.

Results.

- *The spectral formula for a piece* (Theorem 21): for $u = 1$, fixed f, and Riesz means of order $m \geq 2$,

$$\mathcal{P}^{(m)}_u(Y) = c_uY + Y^{1/2}\sum_j\big(\alpha_j(u,f)Y^{it_j} + \overline{\alpha_j(u,f)}Y^{-it_j}\big)\,\gamma_m(t_j) + O(Y^{1/2-\delta}),$$

  the sum over the even Maass cusp forms u_j of $\mathrm{SL}_2(\mathbb{Z})$ with Laplace eigenvalue $\frac{1}{4} + t_j^2$ (and, for $u > 1$, the newforms of the levels dividing u^2), with $\alpha_j(u, f)$ an explicit multiple of the Katok–Sarnak period of u_j at discriminant D, and $\gamma_m(t) \ll |t|^{-m-1}$.

- *Numerical confirmation* (§6): for five quadratics the even parameters explain the variance of $\mathcal{P}_u(Y)/\sqrt{Y}$ at the 92nd–100th percentile among random frequency sets, the odd ones at the 12th–66th; the dominant line $t_1 = 13.7798$ has stable amplitude and phase over two decades of $\log Y$; the two polynomials with exponentially small periods ($D = -8, -163$) show nothing.

- *What the Cesàro conjecture needs* (§7): the sum of the pieces over $u \leq H^{2/3}$ converges to $c_{\mathrm{off}}H + o(H)$ provided the spectral formula holds with an error $O((H/u)^{1/2+\varepsilon}u^A)$, $A < \frac{1}{4}$; the level dependence of the Kuznetsov formula is therefore the whole remaining problem.

2 The piece as a twisted Walfisz sum

Throughout, f is an irreducible quadratic of discriminant D and u is squarefree with $\omega(u) \geq 1$. Call a modulus d *admissible* (for u) if it is squarefree and coprime to $2Du$. Only such d occur: if $p \mid Q_u(h) = u^2h^2 - D$ and $p \nmid 2Du$ then $D \equiv (uh)^2 \pmod p$, so $\big(\frac{D}{p}\big) = 1$ and p is split. The condition "every prime dividing d is split", which appears in part III, is therefore automatic here, and we do not impose it. For admissible d put

$$R_d = R^{(u)}_d = \{x \bmod d :\ u^2x^2 \equiv D\}, \qquad \rho(d) = |R_d| = \omega(d) = 2^{\nu(d)}, \qquad \rho_k(d) = \sum_{x\in R_d} \mathrm{e}\Big(\frac{kx}{d}\Big),$$

ν the number of prime factors and $\mathrm{e}(y) = e^{2\pi iy}$. Two properties are used constantly: R_d is *symmetric*, $x \in R_d \Rightarrow -x \in R_d$, and $0 \notin R_d$, both because $(D, d) = 1$; consequently every $\rho_k(d)$ is real. Write $\lambda(p) = p/(p - 4)$ on split primes, λ multiplicative, and

$$F_u(n) = \sum_{\substack{d\mid n,\ d>1\\ d\text{ admissible}}} \frac{\lambda(d)}{d}, \qquad \mathbb{E}F_u = \sum_{\substack{d>1\\ d\text{ admissible}}} \frac{\lambda(d)\rho(d)}{d^2}, \qquad Q_u(h) = u^2h^2 - D, \tag{1}$$

both sums absolutely convergent ($\lambda(d)\rho(d) \ll d^{\varepsilon}$). *All sums over moduli below run over admissible* $d > 1$: including $d = 1$ would add 1 to F_u and to $\mathbb{E}F_u$, leaving (2) unchanged but spoiling the symmetry used in Lemma 1, since $R_1 = \{0\}$ is not a union of pairs $\{x, -x\}$ with $x \neq -x$. The *sharp piece* and the *Riesz pieces* are

$$S_u(t) = \sum_{h \le t} \big(F_u(Q_u(h)) - \mathbb{E}F_u\big), \qquad \mathcal{P}_u^{(m)}(Y) = \frac{1}{m!} \sum_{h \le Y} (Y-h)^m \big(F_u(Q_u(h)) - \mathbb{E}F_u\big) \quad (m \ge 0), \tag{2}$$

so that $\mathcal{P}_u^{(0)} = S_u$ and, for integers Y, $\mathcal{P}_u^{(1)}(Y) = \sum_{t<Y} S_u(t)$; up to the explicit constant of part III, $\mathcal{P}_u^{(1)}(Y)$ is the piece $\mathcal{P}_u(Y)$ of part III. The connection with part III is that $F_u(Q_u(h))$ is the divisor-sum form of the inner sum of a piece: $d \mid Q_u(h)$ exactly when h lies in one of the $\rho(d)$ classes R_d.

Lemma 1 (sawtooth form)**.** *Let $\psi(y) = \frac{1}{2} - \{y\}$. For every $t \ge 0$,*

$$S_u(t) = \sum_{d \text{ admissible}} \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\Big(\frac{t-x}{d}\Big), \tag{3}$$

the series converging absolutely.

Proof. The right-hand side is not absolutely convergent term by term, since $\sum_d \lambda(d)\rho(d)/d$ diverges; it is summed with the terms x and $-x$ grouped, and converges in that sense, as shown at the end. Expanding F_u and counting,

$$\sum_{h \le t} F_u(Q_u(h)) = \sum_d \frac{\lambda(d)}{d} \#\{h \le t :\ d \mid Q_u(h)\} = \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \#\{h \le t :\ h \equiv x\ (d)\},$$

the interchange being legitimate because $\sum_{d \mid Q_u(h)} \lambda(d)/d$ converges absolutely for each $h \le t$ and there are finitely many h. Represent each $x \in R_d$ by $\langle x \rangle_d \in [1, d]$. For $0 \le t$ and $1 \le x \le d$,

$$\#\{1 \le h \le t :\ h \equiv x\ (d)\} = \Big\lfloor \frac{t-x}{d} \Big\rfloor + 1,$$

valid also when $x > t$, where both sides vanish because $-1 < (t-x)/d < 0$. Writing $\lfloor y \rfloor = y - \frac{1}{2} + \psi(y)$,

$$\#\{h \le t :\ h \equiv x\ (d)\} = \frac{t}{d} + \Big(\frac{1}{2} - \frac{x}{d}\Big) + \psi\Big(\frac{t-x}{d}\Big).$$

Now sum over $x \in R_d$. Since R_d is symmetric and $0 \notin R_d$, its representatives split into pairs $\{\langle x \rangle_d,\ d - \langle x \rangle_d\}$, and $(\frac{1}{2} - \frac{x}{d}) + (\frac{1}{2} - \frac{d-x}{d}) = 0$; hence $\sum_{x \in R_d} (\frac{1}{2} - \langle x \rangle_d / d) = 0$ for every admissible $d > 1$. Therefore

$$\sum_{h \le t} F_u(Q_u(h)) = t \sum_d \frac{\lambda(d)\rho(d)}{d^2} + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\Big(\frac{t-x}{d}\Big) = t\, \mathbb{E}F_u + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\Big(\frac{t-x}{d}\Big),$$

which is (3). For the convergence, group the terms of a pair $\{x, -x\}$ and write $a = \langle x \rangle_d$ with $a \le d/2$, so that the pair is $\{a, d-a\}$. If $d > 2t$ and $a > t$ then $\{(t-a)/d\} = (t-a+d)/d$ and $\{(t+a)/d\} = (t+a)/d$, so by the periodicity of ψ the two sawtooths sum to $-2t/d$; if $a \le t$ their sum is bounded by 1, and this happens for at most $\rho(d)$ values with $a \le t < d/2$, a set that is empty once $d > 2t$. Hence for $d > 2t$ each pair contributes $\ll t/d$ and

$$\sum_{d > 2t} \frac{\lambda(d)}{d} \cdot \frac{t\,\rho(d)}{d} \ \ll\ t \sum_{d > 2t} \frac{\lambda(d)\rho(d)}{d^2} \ \ll\ t^{\varepsilon},$$

while the range $d \le 2t$ is a finite sum. So the paired series converges and (3) holds. $\square$

In words. A piece counts, for each admissible modulus d, how far the number of $h \le t$ with $d \mid Q_u(h)$ departs from its expectation, weighted by $\lambda(d)/d$; and that departure is a sawtooth evaluated at the roots. The linear term that normally accompanies such a count cancels because the roots come in pairs $\pm x$. This is the same cancellation that makes the diagonal of part I a sum of sawtooths $\{H/d\} - \frac{1}{2}$, and it is why the piece is a Walfisz-type sum: in the classical case $\sum_{n \le x} n^{-1}(\frac{1}{2} - \{x/n\})$ one sums a sawtooth over all moduli with weight $1/n$, and here one sums it over admissible moduli, at the roots of a quadratic congruence, with weight $\lambda(d)/d$.

Lemma 2 (Weyl-sum form)**.** *For every $t \ge 0$ and every $K \ge 1$,*

$$S_u(t) = \frac{1}{\pi} \sum_{k \ge 1} \frac{1}{k} \sum_{d \text{ admissible}} \frac{\lambda(d)\rho_k(d)}{d} \sin \frac{2\pi k t}{d}, \tag{4}$$

the k-sum converging boundedly. Consequently, for the Riesz pieces of order $m \ge 1$ and integer Y,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{\pi} \sum_{k \ge 1} \frac{1}{k} \sum_{d} \frac{\lambda(d)\rho_k(d)}{d} \Phi_m\Big(\frac{2\pi k}{d}; Y\Big), \qquad \Phi_m(\theta; Y) = \frac{1}{m!} \sum_{h \le Y} (Y-h)^m \sin(\theta h), \tag{5}$$

with $\Phi_m(\theta; Y) \ll_m \min\big(Y^{m+1},\ Y^m \|\theta/2\pi\|^{-1}\big)$, $\|\cdot\|$ the distance to $\mathbb{Z}$.

Proof. The Fourier expansion $\psi(y) = \sum_{k \ge 1} \sin(2\pi k y)/(\pi k)$ holds for $y \notin \mathbb{Z}$, boundedly and uniformly on compact subsets of $\mathbb{R} \setminus \mathbb{Z}$, and the partial sums are uniformly bounded. Insert it into (3) and use, for each d,

$$\sum_{x \in R_d} \sin \frac{2\pi k(t-x)}{d} = \Im\Big[\mathrm{e}\Big(\frac{kt}{d}\Big)\overline{\rho_k(d)}\Big] = \rho_k(d) \sin \frac{2\pi k t}{d},$$

since $\rho_k(d)$ is real. The interchange of the k-sum with the d-sum is justified by the uniform boundedness of the partial sums of the Fourier series together with the absolute convergence established in Lemma 1 (dominated convergence for series). For (5), apply $\frac{1}{m!} \sum_{h \le Y} (Y-h)^m$ to $t = h$ in (4), which is a finite sum, and evaluate Φ_m by summation by parts m times against the geometric sum $\sum_{h \le Y} \mathrm{e}(\theta h/2\pi) \ll \min(Y, \|\theta/2\pi\|^{-1})$. □

Remark 3. Two features of (5) decide everything that follows. First, the frequencies k enter only through the *Weyl sums* $\rho_k(d)$ of the dilated roots, so the arithmetic of the problem is entirely contained in the Dirichlet series

$$W_k(s) = \sum_{d \text{ admissible}} \frac{\lambda(d)\rho_k(d)}{d^s} \qquad (\Re s > 1), \tag{6}$$

one for each k. Second, the weight $1/k$ in front decays only like k^{-1}, so a bound for W_k that loses a power of k is expensive: this is the same phenomenon that forced the condition $\theta + 6B < 1$ in part III, Theorem A, and it is the reason we do not bound the $\rho_k(d)$ one frequency at a time. The route of this paper is instead to continue $W_k(s)$ analytically, where the dependence on k appears as a Fourier coefficient of an automorphic form and therefore *decays* on average.

Proposition 4 (the Mellin representation)**.** *Let $m \ge 1$ and let*

$$A_u(s) = \sum_{h \ge 1} \frac{F_u(Q_u(h)) - \mathbb{E}F_u}{h^s} \qquad (\Re s > 1).$$

Then A_u extends to a function holomorphic in $\Re s > \frac{1}{2}$ except possibly for the singularities inherited from the W_k, and for integer $Y \ge 2$ and any $c > 1$,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{2\pi i} \int_{(c)} A_u(s) \frac{Y^{s+m}}{s(s+1)\cdots(s+m)} \, ds. \tag{7}$$

Moreover, for $0 < \Re s < 1$,

$$A_u(s) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sin\frac{\pi s}{2} \sum_{k\geq 1} k^{s-1}\, W_k(s+1) \;+\; \big(\textit{entire}\big), \tag{8}$$

whenever the k*-sum converges, where* W_k *is as in* (6).

Proof. (7) is the Riesz–Perron formula, valid because $F_u(Q_u(h)) - \mathbb{E}F_u \ll h^{\varepsilon}$ so that A_u converges absolutely for $\Re s > 1$. For (8), expand F_u and sum over each residue class:

$$\sum_{h\geq 1} \frac{F_u(Q_u(h))}{h^s} = \sum_d \frac{\lambda(d)}{d^{1+s}} \sum_{x\in R_d} \zeta\Big(s, \frac{\langle x\rangle_d}{d}\Big),$$

$\zeta(s,\alpha)$ the Hurwitz zeta function. Since $\zeta(s,\alpha)$ has residue 1 at $s=1$ for every α, the pole of the d-th term is $\rho(d)\lambda(d)d^{-1-s}$, and summing over d these residues total $\mathbb{E}F_u$, which is exactly the pole of $\mathbb{E}F_u\zeta(s)$: each bracket $\lambda(d)d^{-1-s}\sum_x \zeta(s,\langle x\rangle_d/d) - \lambda(d)\rho(d)d^{-2}\zeta(s)$ is entire, which is the statement that A_u continues past $s=1$. Hurwitz's formula,

$$\zeta(s,\alpha) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\Big[\sin\frac{\pi s}{2}\sum_{k\geq 1}\frac{\cos 2\pi k\alpha}{k^{1-s}} + \cos\frac{\pi s}{2}\sum_{k\geq 1}\frac{\sin 2\pi k\alpha}{k^{1-s}}\Big],$$

valid for $\Re s < 0$ and by continuation for $\Re s < 1$ with the series interpreted as the analytic continuations of the periodic zeta functions, gives after summing over the symmetric set R_d

$$\sum_{x\in R_d}\zeta\Big(s,\frac{\langle x\rangle_d}{d}\Big) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\sin\frac{\pi s}{2}\sum_{k\geq 1}\frac{\rho_k(d)}{k^{1-s}},$$

the sine term vanishing identically because $\sum_{x\in R_d}\sin(2\pi kx/d) = 0$ by symmetry. Multiplying by $\lambda(d)d^{-1-s}$ and summing over d gives (8). □

In words. The piece is determined by one Dirichlet series for each frequency, namely the Dirichlet series of the Weyl sums of the roots of $u^2x^2 \equiv D$; the symmetry of the root set kills the odd half of Hurwitz's formula, so only the even half survives. That the odd half disappears here is the first appearance of the parity phenomenon that will select the even Maass forms in §4, and it has the same source: the root set of a quadratic congruence is invariant under $x \mapsto -x$.

3 Salié sums and the Dirichlet series of the Weyl sums

By Remark 3 everything now depends on the series $W_k(s) = \sum_d \lambda(d)\rho_k(d)d^{-s}$. This section identifies its coefficients as Kloosterman sums of half-integral weight, which is what makes an analytic continuation possible. Write, for $c \geq 1$ odd with $(D,c)=1$ and $h\in\mathbb{Z}$,

$$\mathcal{W}_h(D;c) = \sum_{b \bmod c,\ b^2\equiv D} \mathrm{e}\Big(\frac{hb}{c}\Big), \tag{9}$$

the Weyl sum of the roots of the *fixed* quadratic congruence, and let $K_\chi(m,n;p) = \sum^*_{x \bmod p}\big(\frac{x}{p}\big)\mathrm{e}\big(\frac{mx+n\bar{x}}{p}\big)$ be the Kloosterman sum twisted by the Legendre symbol, $\varepsilon_p = 1$ or i according as $p\equiv 1$ or 3 (mod 4).

Lemma 5 (the dilation is a change of frequency)**.** *For admissible* d *and* $k\geq 1$*, with* $\bar{u}$ *the inverse of* u *modulo* d*,*

$$\rho_k(d) = \mathcal{W}_{k\bar{u}}(D;d).$$

Proof. $x \mapsto b = ux$ is a bijection of $R_d^{(u)} = \{x : u^2x^2 \equiv D\}$ onto $\{b : b^2 \equiv D\}$ because $(u,d) = 1$, and $kx \equiv k\bar{u}b \pmod{d}$. □

So the dilation by u does not change the quadratic; it moves the frequency from k to $k\bar{u}$, a frequency that depends on the modulus. This is the exact point at which the pieces with $u > 1$ leave the range of the theorems of Duke, Friedlander and Iwaniec, which are stated for frequencies fixed independently of the modulus.

Lemma 6 (Salié). *Let p be an odd prime with $\left(\frac{D}{p}\right) = 1$ and let $h \in \mathbb{Z}$ with $p \nmid h$. Then*

$$\mathcal{W}_{2h}(D;p) = \frac{1}{\varepsilon_p\sqrt{p}}\, K_\chi\big(D, h^2; p\big). \tag{10}$$

For odd squarefree $c = c_1c_2$ with $(c_1, c_2) = 1$ and $(D, c) = 1$,

$$\mathcal{W}_h(D;c) = \mathcal{W}_{h\overline{c_2}}(D;c_1)\,\mathcal{W}_{h\overline{c_1}}(D;c_2), \tag{11}$$

so (10) determines $\mathcal{W}_h(D;c)$ for every odd squarefree c coprime to $2D$.

Proof. (11) is the Chinese remainder theorem: $b \mapsto (b \bmod c_1, b \bmod c_2)$ is a bijection of the roots modulo c onto the pairs of roots, and $b/c \equiv b_1\overline{c_2}/c_1 + b_2\overline{c_1}/c_2 \pmod{1}$. For (10), substitute $b = 2\bar{h}y$ in (9), which is a bijection of the roots of $b^2 \equiv D$ onto the roots of $y^2 \equiv h^2D\bar{4}$ and turns $\mathrm{e}(2hb/p)$ into $\mathrm{e}(2y/p)$; hence $\mathcal{W}_{2h}(D;p) = \sum_{y^2\equiv h^2D\bar{4}} \mathrm{e}(2y/p)$. Salié's evaluation of the twisted Kloosterman sum, $K_\chi(m,n;p) = \varepsilon_p\sqrt{p}\left(\frac{m}{p}\right)\sum_{y^2\equiv mn\,(p)} \mathrm{e}\big(\frac{2y}{p}\big)$ [6, §4.5], applied with $m = D$, $n = h^2$ gives (10), the factor $\left(\frac{D}{p}\right)$ being 1 because p is split. [1] □

Remark 7 (what Lemma 6 does and does not give). Combining Lemmas 5 and 6, for even k and admissible d the coefficient of $W_k(s)$ is, up to the elementary factor $\varepsilon_d^{-1}d^{-1/2}$, a Kloosterman sum of half-integral weight with the two arguments D and $(k\bar{u}/2)^2$: the first fixed by the polynomial, the second carrying the frequency and the dilation. This is the shape to which the Kuznetsov formula of weight one half applies, and it is the machinery behind every known estimate for these Weyl sums *on average over the modulus* [5, 2, 3].

It is, however, *not* the route to the oscillations observed in §6, and it is worth saying why, because the distinction is easy to miss. Selberg's Kloosterman zeta function is normalised as $Z_{m,n}(\sigma) = \sum_c S(m,n;c)c^{-2\sigma}$, with poles at $\sigma = s_j$, $\lambda_j = s_j(1-s_j)$ [4, Theorem 1], and their Remark 3 covers exactly our case, weight $\frac{1}{2}$ on $\Gamma_0(4N)$. Tracing that normalisation through Lemma 6 and Proposition 4 places the singularities of A_u at $s = -\frac{1}{2} + 2it_j$ and predicts an oscillation of $\mathcal{P}_u^{(1)}(Y)/\sqrt{Y}$ at frequency $2t_j$. That prediction is false: fitted at the frequencies $2t_j$ the even set explains between 2.1% and 2.2% of the variance and sits at the 43rd to 69th percentile of random frequency sets drawn from the same band, against the 99.7th to 100th percentile at the frequencies t_j (§6; `scripts/piece-freq.ts`). The reason for the mismatch is that $Z_{m,n}$ sums over the *modulus*, whereas $\mathcal{P}_u$ sums over the *argument* h. The two are different analytic problems, and only the second produces the observed law.

The argument-side expansion. The route that does fit is Bykovskiĭ's [1]: write the sum over $h \le Y$ as a finite combination of Poincaré series for $\mathrm{SL}_2(\mathbb{Z})$ and apply the spectral decomposition in weight 0 and level 1. There the parametrisation $\lambda_j = s_j(1-s_j)$, $s_j = \frac{1}{2} + it_j$, produces terms $Y^{s_j} = \sqrt{Y}\,Y^{it_j}$ directly, exactly as the error term in the hyperbolic lattice-point problem does. This accounts at once for all four features of the data: the frequencies are the t_j themselves, the scale is $\sqrt{Y}$, the spectrum is that of level 1 and weight 0, and the forms that can appear

[1] Both identities were verified numerically for all split $p \le 37$, all $D \in \{-4, -3, 5, 8, 12\}$ and $h \in \{1, 2, 3, 5\}$, and (11) for composite c; the reference for Salié's evaluation should be pinned to a precise statement of the normalisation.

are those pairing non-trivially with the Heegner points or the closed geodesics of discriminant D, which by Proposition 9 are the even ones. What we shall prove is the following; the proof occupies §§5–5.2, and the precise form, with the order of the Riesz mean, is Theorem 21.

Theorem 8 (the spectral formula for a piece)**.** *Let $u = 1$ and let $\mathcal{G}$ be the model object (24), smoothed by a Riesz mean of sufficiently high order. There are coefficients $\alpha_j(D)$, supported on the even Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$ with $\lambda_j = \frac{1}{4} + t_j^2$ and proportional to the Katok–Sarnak periods $\mathrm{Per}_D(u_j)$, such that*

$$\mathcal{G}(Y) = \sqrt{Y} \sum_j \big(\alpha_j(D) Y^{it_j} + \overline{\alpha_j(D)} Y^{-it_j}\big) + \mathcal{E}(Y) + O\big(Y^{1/2-\delta}\big)$$

for some $\delta > 0$, the sum converging absolutely and $\mathcal{E}$ being the contribution of the continuous spectrum, which is $o(\sqrt{Y})$.

For fixed D this is in substance the content of Bykovskiĭ's spectral treatment of the divisor sums $\sum_{|n|\le P} \sigma_{-s}(n^2 + D)$ [1], which is where an asymptotic of this shape was first obtained; what we require beyond it is the identification of the oscillating terms, their amplitudes and the error exponent. The proof given in §5.2 does not go through the Weyl-sum series W_k at all; it works with the test function directly, which is what makes it elementary. Lemmas 5 and 6 are retained because the identification of the Weyl sums as Salié sums is what any *quantitative* treatment of the remainder, or any attempt at the uniformity in u of §7, will have to use.

4 Katok–Sarnak and the parity rule

The route of §3 would place the poles of W_k at $\frac{1}{2} \pm it_j$, t_j running over the weight-$\frac{1}{2}$ spectrum, with residues proportional to $\overline{\rho_j(|D|)}\rho_j((k/2)^2)$. Two classical inputs turn this into a statement about the Maass forms of $\mathrm{SL}_2(\mathbb{Z})$.

Shimura and Katok–Sarnak. The Shimura correspondence sends the weight-$\frac{1}{2}$ forms of level 4 in Kohnen's plus space to weight-0 Maass forms of level 1 with the same spectral parameter, and the image consists of the *even* forms; the odd Maass forms correspond to weight $\frac{3}{2}$. Katok and Sarnak [7] identify the Fourier coefficients of the weight-$\frac{1}{2}$ form at a fundamental discriminant D with a period of the corresponding Maass form u_j: for $D < 0$ the sum of u_j over the Heegner points of discriminant D, and for $D > 0$ the integral of u_j over the closed geodesics of discriminant D. Writing $\mathrm{Per}_D(u_j)$ for that period,

$$\rho_j(|D|) = \kappa(t_j, D)\, \mathrm{Per}_D(u_j) \tag{12}$$

with an explicit factor κ. [2]

Proposition 9 (parity)**.** *If u_j is an odd Maass form then $\mathrm{Per}_D(u_j) = 0$, for every discriminant D. Consequently only the even Maass forms of $\mathrm{SL}_2(\mathbb{Z})$ can contribute to the pieces.*

Proof. Let $\iota(z) = -\bar{z}$, the reflection in the imaginary axis; u_j is odd exactly when $u_j \circ \iota = -u_j$. For $D < 0$ the Heegner points of discriminant D are the roots $z_Q \in \mathbb{H}$ of the integral binary quadratic forms $Q = [a, b, c]$ of discriminant D, one for each class, and $\iota(z_{[a,b,c]}) = z_{[a,-b,c]}$; since $[a, b, c] \mapsto [a, -b, c]$ is the inversion of the class group, it permutes the classes, so ι permutes the set of Heegner points. Hence $\mathrm{Per}_D(u_j) = \sum_Q u_j(z_Q) = \sum_Q u_j(\iota z_Q) = -\mathrm{Per}_D(u_j)$, which forces $\mathrm{Per}_D(u_j) = 0$. For $D > 0$ the same involution permutes the closed geodesics S_Q attached to the classes, and reverses none of the identifications used in defining the cycle integral, so the same computation applies to $\mathrm{Per}_D(u_j) = \sum_Q \int_{\Gamma_Q \backslash S_Q} u_j \, ds$. □

[2] Only $|\kappa|$, and only its dependence on D, is used below; the normalisation of Per should be pinned to and the normalisation of Per to [7]; only $|\kappa|$, and only its dependence on D, is used below.

In words. The Heegner points of a given discriminant, and likewise the closed geodesics, are symmetric about the imaginary axis; an odd Maass form is antisymmetric about it; so its period vanishes. This is the third appearance of one and the same symmetry. In §2 it made the linear term of the sawtooth identity vanish; in Proposition 4 it killed the odd half of Hurwitz's formula; here it removes the odd half of the spectrum. All three are the statement that the roots of a quadratic congruence come in pairs $\pm x$.

Corollary 10 (the shape of a piece)**.** *In Theorem 8 the spectral sum runs over the even Maass cusp forms of* $\mathrm{SL}_2(\mathbb{Z})$ *only, and the coefficient of the line* t_j *is proportional to* $\mathrm{Per}_D(u_j)$*.*

The size of the periods, and a visibility threshold. For $D < 0$ of class number one the period is the single value $u_1(z_D)$ at $z_D = (-b + i\sqrt{|D|})/2$. Through the Fourier expansion $u_j(z) = 2\sqrt{y}\sum_{n\geq 1} a_j(n)K_{it_j}(2\pi ny)\cos(2\pi nx)$ this is governed by $K_{it_j}(2\pi y)$ with $y = \sqrt{|D|}/2$. The Bessel function $K_{it}(x)$ of imaginary order is of size $e^{-\pi t/2}$ and oscillatory while $x < t$, and only begins to decay in x once $x > t$. Hence the periods of a fixed form u_j are all of comparable size for

$$|D| < \left(\frac{t_j}{\pi}\right)^2, \tag{13}$$

and collapse exponentially beyond it. For the first even form, $t_1 = 13.7798$, the threshold is $|D| \approx 19.2$. This is a falsifiable prediction: the first line should be visible for the small discriminants and invisible for $|D| \geq 43$, while the higher lines, having larger t_j, should persist further. Section 6 reports what is observed.

5 The spectral formula for a piece

We now describe the argument-side route in the form we believe it takes, for $u = 1$ and the model object (24), and say precisely which steps are established and which are not. Throughout $D < 0$; for $D > 0$ every occurrence of a Heegner point is replaced by a closed geodesic and the argument is parallel.

Step 1: the root pairs are Heegner points. Let $\mathcal{Q}_D$ be the set of integral binary quadratic forms of discriminant D and, for $Q = [a, b, c] \in \mathcal{Q}_D$ with $a > 0$, let $z_Q = (-b + \sqrt{D})/(2a) \in \mathbb{H}$. Gauss's parametrisation, in the form used by Hooley [5, §3] and studied in general by Welsh [8], is the bijection

$$\{(d, b) :\ d \geq 1,\ b \bmod 2d,\ b^2 \equiv D\ (4d)\} \longleftrightarrow \{Q = [d, b, *] \in \mathcal{Q}_D\}, \qquad (d, b) \mapsto z_{(d,b)} = \frac{-b + \sqrt{D}}{2d}, \tag{14}$$

under which the pairs (d, b) become the Heegner points of discriminant D, falling into exactly $h(D)$ orbits of $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$. [3] The two coordinates of a Heegner point carry exactly the two data of a root pair:

$$\Im z_{(d,b)} = \frac{\sqrt{|D|}}{2d}, \qquad \Re z_{(d,b)} = -\frac{b}{2d}. \tag{15}$$

Step 2: the piece is a sum over a Heegner orbit. Substituting (15) into the sawtooth form of Lemma 1 turns the sum over root pairs into a sum over Heegner points:

$$S(t) = \frac{2}{\sqrt{|D|}}\sum_{z} \Phi_t(z), \qquad \Phi_t(z) = \Im z \cdot \psi\Big(\frac{2t\,\Im z}{\sqrt{|D|}} + 2\Re z\Big), \tag{16}$$

[3]We verified (14) numerically: reducing $z_{(d,b)}$ to the fundamental domain for all $d \leq 60$ gives exactly $h(D)$ distinct points, for $D = -3, -4, -8, -11, -20, -23, -24$ with $h = 1, 1, 1, 1, 2, 3, 2$. The conventions relating $b^2 \equiv D\ (4d)$ to the $b^2 \equiv D\ (d)$ of §2 must be fixed carefully in the final version.

the sum over all Heegner points of discriminant D, i.e. over the $h(D)$ orbits Γz_Q. This is the whole content of the change of route: the object is not a sum over moduli carrying Kloosterman sums, but a sum of a fixed test function over an orbit of Γ in $\mathbb{H}$, with the length t entering only as a dilation parameter of the test function.

Remark 11 (the Weyl series is a Poincaré-type series). Before going on, it is worth recording what (16) says about the Dirichlet series of the Weyl sums themselves, since it is the reason the spectrum can enter at all. Directly from the coordinates (15), for $\Re s > 1$,

$$\sum_{z\ \text{Heegner}} (\Im z)^s\, \mathrm{e}(m\,\Re z) = \Big(\frac{\sqrt{|D|}}{2}\Big)^s \sum_{d\geq 1} \frac{1}{d^s} \sum_{b^2\equiv D\,(4d)} \mathrm{e}\Big(\frac{-mb}{2d}\Big) : \tag{17}$$

the Dirichlet series of the Weyl sums of a quadratic congruence is a Poincaré-type series summed over the Heegner points of that discriminant. [4] The proof below does not use (17); it works with the test function directly, which is what makes it elementary. We record the identity because it is the structural statement behind everything, and because §6.1 tests the same law on the bare sums it describes.

Step 3: unfolding and the period. One point of bookkeeping decides the shape of everything that follows. The arithmetic parametrisation of (14) is by pairs (d, b) with b taken *modulo* $2d$, and $b \mapsto b + 2d$ is $z \mapsto z - 1$; so the sum in (16) runs over the Heegner points *modulo translation*, that is over $\Gamma_\infty\backslash\Gamma z_Q$ and not over Γz_Q. This is not a normalisation: a full Γ-orbit contains $z_Q + n$ for every $n \in \mathbb{Z}$, all of the same height, and $\sum_{\gamma\in\Gamma}(\Im\gamma w)^2$ diverges for that reason, whereas $\sum_{\gamma\in\Gamma_\infty\backslash\Gamma}(\Im\gamma w)^2 = E(w, 2)$ converges. Accordingly we set

$$P[\Phi_t](w) = \sum_{\gamma\in\Gamma_\infty\backslash\Gamma} \Phi_t(\gamma w),$$

a Poincaré series in the classical sense, so that $S(t) = \frac{2}{\sqrt{|D|}} \sum_{Q\in\Gamma\backslash\mathcal{Q}_D} |\Gamma_{z_Q}|^{-1} P[\Phi_t](z_Q)$. The spectral decomposition of $P[\Phi_t]$ on $L^2(\Gamma\backslash\mathbb{H})$, together with the classical unfolding $\langle P[\Phi_t], u_j\rangle = \int_0^\infty\!\int_0^1 \Phi_t(z)\overline{u_j(z)}\,\frac{dx\,dy}{y^2}$ — the x-integral over a single period, which is exactly the integral computed in Step 4 — gives

$$S(t) = \frac{2}{\sqrt{|D|}}\Big[(\text{constant and Eisenstein terms}) + \sum_j \langle\Phi_t, u_j\rangle \sum_{Q\in\Gamma\backslash\mathcal{Q}_D} \frac{u_j(z_Q)}{|\Gamma_{z_Q}|}\Big], \tag{18}$$

in which the geometric factor is precisely the Katok–Sarnak period $\mathrm{Per}_D(u_j)$ of §4. By Proposition 9 it vanishes for every odd u_j: the parity rule is now visible as the statement that the Heegner orbit is symmetric about the imaginary axis while an odd form is antisymmetric.

Step 4: where the frequencies come from. No continuation theorem is needed here. Expand ψ in Fourier series; since ψ is odd,

$$\psi(a + 2x) + \psi(a - 2x) = \frac{2}{\pi}\sum_{k\geq 1} \frac{\sin(2\pi ka)\cos(4\pi kx)}{k}, \qquad a = \frac{2t\,\Im z}{\sqrt{|D|}},$$

so that $\Phi_t^+(z) = \frac{y}{\pi}\sum_k k^{-1}\sin(2\pi ka)\cos(4\pi kx)$. Pair this against the expansion of an even Maass form, $u_j(z) = 2\sqrt{y}\sum_{n\geq 1} a_j(n) K_{it_j}(2\pi ny)\cos(2\pi nx)$. The x-integral is $\int_0^1 \cos(4\pi kx)\cos(2\pi nx)\,dx = \frac{1}{2}\delta_{n,2k}$, so only the coefficients at even indices survive and

$$\langle\Phi_t^+, u_j\rangle = \sum_{k\geq 1} \frac{a_j(2k)}{\pi k} I_k, \qquad I_k = \int_0^\infty y^{-1/2}\sin(\beta_k y)\, K_{it_j}(\alpha_k y)\, dy, \quad \alpha_k = 4\pi k,\ \beta_k = \frac{4\pi kt}{\sqrt{|D|}}. \tag{19}$$

[4] Verified numerically for $D = -4$, $s = 1.6$, $m = 1$ to 10^{-16}.

Substituting $y = w/\beta_k$ gives $I_k = \beta_k^{-1/2} \int_0^\infty w^{-1/2} \sin(w) K_{it_j}(\alpha_k w/\beta_k)\, dw$, in which $\alpha_k/\beta_k = \sqrt{|D|}/t$ is small. Two standard facts finish the computation: the small-argument expansion

$$K_{i\tau}(x) = \tfrac{1}{2}\Big[\Gamma(-i\tau)\big(\tfrac{x}{2}\big)^{i\tau} + \Gamma(i\tau)\big(\tfrac{x}{2}\big)^{-i\tau}\Big] + O(x^2), \tag{20}$$

and $\int_0^\infty w^{\sigma-1} \sin w\, dw = \Gamma(\sigma)\sin(\pi\sigma/2)$ for $0 < \Re\sigma < 1$, applied at $\sigma = \frac{1}{2} \pm it_j$. Hence

$$I_k = \beta_k^{-1/2}\Big[\Gamma(-it_j)\Gamma\big(\tfrac{1}{2}+it_j\big)\sin\tfrac{\pi(\frac{1}{2}+it_j)}{2}\Big(\frac{\sqrt{|D|}}{2t}\Big)^{it_j}\cdot\tfrac{1}{2} + \overline{(\,\cdot\,)}\Big] + O\big((\sqrt{|D|}/t)^2\big). \tag{21}$$

Every t-dependence is now explicit: $\beta_k^{-1/2} \propto t^{-1/2}$ and the bracket carries $t^{\mp it_j}$. So $\langle \Phi_t^+, u_j\rangle = t^{-1/2}\big(A_j(D)\, t^{-it_j} + \overline{A_j(D)}\, t^{it_j}\big) + O(t^{-5/2})$, and by (18) the same holds for $S(t)$ with A_j multiplied by $\mathrm{Per}_D(u_j)$. Summing over $t < Y$ turns $t^{-1/2\mp it_j}$ into $Y^{1/2\mp it_j}$, which is the observed law.

Two features of the data are read off (21) directly. The frequencies are the t_j themselves, not $2t_j$, because (20) produces $x^{\pm it_j}$ and not $x^{\pm 2it_j}$. And the phase carries the factor $(\sqrt{|D|}/2)^{it_j}$, so that changing the discriminant shifts the phase by exactly $\frac{t_j}{2}\log|D|$ plus the sign of $\mathrm{Per}_D(u_j)$: this is precisely the prediction (25) tested in Tables 4 and 5, and it is what those tables confirm to a median of 0.012 radians. [5]

Remark 12 (what this route does *not* need). The derivation above uses no continuation theorem for any zeta or Poincaré series: only the Fourier expansions of ψ and of u_j, the small-argument behaviour of the Bessel function, and one elementary integral. This matters, because our two earlier attempts to route the argument through such a theorem both failed. The first, through Selberg's Kloosterman zeta function, predicted the frequencies $2t_j$ and was refuted by the data (Remark 7). The second, through the meromorphic continuation of the Poincaré series of [4], does not apply: their $P_m(z,s)$ carries the factor $\mathrm{e}(m\gamma z)$, whose imaginary part contributes an exponential decay $e^{-2\pi m\Im\gamma z}$ and puts the series in L^2, whereas the series arising in (17) carries only $\mathrm{e}(m\,\Re\gamma z)$ and is not the same object. The identity (17) remains true and remains the reason the spectrum enters; what is not available is a citable continuation of the particular series it produces.

The same computation bears on the size of these sums. Formally, (21) makes every spectral term of $S(t)$ of size $t^{-1/2}$, so the Riesz mean is $O(Y^{1/2+\varepsilon})$ as soon as the sum over j converges: square-root cancellation, against the $X^{3/4+\varepsilon}$ of Hooley [5] and the $X^{2/3+\varepsilon}$ of Bykovskiĭ [1] for the sharp sums. That is exactly what §6.1 observes, with $T_1(X)/\sqrt{X}$ below 0.56, 0.53 and 0.86 for $D = -4, -3, -8$ over $X \le 10^6$. We do not state it as a proposition, because the convergence of the sum over j is the unfinished step below and there is no shortcut round it: an earlier version of this paper deduced it instead from a continuation theorem that, on inspection, applies to a different series (Remark 12).

Theorem 8 is proved, in the precise form of Theorem 21, in §5.2.

Step 5: why the Poincaré series converges. As it stands $|\Phi_t(z)| \le \frac{1}{2}\Im z$ and $\sum_{\gamma\in\Gamma_\infty\backslash\Gamma}\Im(\gamma w)$ diverges, so $P[\Phi_t]$ is not absolutely convergent as it stands. The divergence is exactly that of the Eisenstein series at $s = 1$. It is removed by the symmetry of the Heegner orbit, which we may as well build into the seed.

Lemma 13 (the symmetrised seed decays quadratically)**.** *Write $z = x + iy$, $a = a_t(y) = 2ty/\sqrt{|D|}$, and $\Phi_t^+(z) = \frac{1}{2}\big(\Phi_t(z) + \Phi_t(-\bar z)\big) = \frac{y}{2}\big(\psi(a+2x) + \psi(a-2x)\big)$. Then*

(i) if $\|2x\| > a$, where $\|\cdot\|$ is the distance to $\mathbb{Z}$ and $a < \frac{1}{2}$, then $\Phi_t^+(z) = -ay = -2ty^2/\sqrt{|D|}$ exactly;

[5] Both standard facts were checked numerically; (20) to a relative 10^{-6} at $x \le 0.1$ for $t = 13.78, 17.74$.

(ii) $|\Phi_t^+(z)| \le y/2$ *always, and the exceptional set* $\{x : \|2x\| \le a\}$ *has measure* $2a$ *in* x *modulo* 1*.*

Moreover Φ_t^+ *may replace* Φ_t *in* (16) *without changing the value, because the Heegner orbit is stable under* $z \mapsto -\bar{z}$*: the pair* (d, b) *maps to* $(d, -b)$*.*

Proof. ψ is odd and, on any interval free of integers, satisfies $\psi(\theta + a) = \psi(\theta) - a$. If $\|2x\| > a$ then neither $2x$ nor $-2x$ crosses an integer when shifted by a, so $\psi(2x + a) + \psi(-2x + a) = \psi(2x) + \psi(-2x) - 2a = -2a$, giving (i); (ii) is $|\psi| \le \frac{1}{2}$ and the definition of the exceptional set. The last statement is $z_{(d,-b)} = -\overline{z_{(d,b)}}$ for $D < 0$. □

So the seed is $O(t\,y^2)$ off a set of x-measure $O(ty)$, on which it is $O(y)$; both contributions to $\sum_{\gamma\in\Gamma_\infty\backslash\Gamma} |\Phi_t^+(\gamma w)|$ are then dominated by $t \sum_{\gamma\in\Gamma_\infty\backslash\Gamma} (\Im\gamma w)^2 = tE(w, 2)$, which converges. The arithmetic form of this convergence is the tail estimate already proved in Lemma 1. This is the fourth and last appearance of one symmetry: the root set of a quadratic congruence is stable under $x \mapsto -x$, and that single fact makes the linear term of the sawtooth identity vanish (Lemma 1), kills the odd half of Hurwitz's formula (Proposition 4), annihilates the periods of the odd Maass forms (Proposition 9), and now renders the Poincaré series convergent.

5.1 Completing the decomposition

Three things were left open above: the two conventions for the modulus, the sense in which the spectral decomposition of Step 3 is legitimate, and the order of Riesz mean at which the sum over the spectrum converges. We settle all three.

Lemma 14 (the two conventions agree)**.** *Let* d *be odd and coprime to* $2D$*. Reduction modulo* d *is a bijection*

$$\{b' \bmod 2d : \ b'^2 \equiv D\ (4d)\} \ \longrightarrow\ \{b \bmod d : \ b^2 \equiv D\ (d)\},$$

and under it $\mathrm{e}\big(2k\,b'/(2d)\big) = \mathrm{e}(kb/d)$ *for every* k*. Hence the frequency* k *of the sawtooth expansion of §2 corresponds to the frequency* $m = 2k$ *of* (17)*, and the two parametrisations describe the same sums.*

Proof. Since $(d, 2) = 1$, the Chinese remainder theorem splits the condition $b'^2 \equiv D\ (4d)$ into $b'^2 \equiv D\ (d)$ and $b'^2 \equiv D\ (4)$. The latter is solvable because $D \equiv 0$ or $1 \pmod 4$, and it determines b' modulo 2; combined with b' modulo d this determines b' modulo $2d$. So each b has exactly one lift, and the map is a bijection. The phases agree because $b' \equiv b \pmod d$. [6] □

Lemma 15 (truncation, and why the decomposition is legitimate)**.** *Every Heegner point of discriminant* D *satisfies* $\Im z = \sqrt{|D|}/(2d) \le \sqrt{|D|}/2$*. Consequently, replacing* Φ_t^+ *by its truncation* $\Phi_t^\flat = \Phi_t^+ \cdot \mathbf{1}_{\{\Im z \le \sqrt{|D|}/2\}}$ *changes neither side of* (16)*. The truncated seed satisfies*

(i) $|\Phi_t^\flat(z)| \le \sqrt{|D|}/4$*, and* $\Phi_t^\flat(z) = 0$ *for* $\Im z > \sqrt{|D|}/2$*;*

(ii) $\Phi_t^\flat(z) \ll t\,(\Im z)^2$ *as* $\Im z \to 0$*, in the sense of Lemma 13;*

(iii) $\int_0^1 \Phi_t^\flat(x + iy)\,dx = 0$ *for every* y*.*

Hence $P[\Phi_t^\flat]$ *is bounded on* $\Gamma\backslash\mathbb{H}$*, lies in* L^2*, and its spectral decomposition is valid.*

Proof. (i) and (ii) are Lemma 13 and the truncation. (iii) holds because ψ has mean zero and $x \mapsto 2x$ covers two full periods, so $\int_0^1 [\psi(a+2x)+\psi(a-2x)]dx = 0$ for every y; the truncation does not depend on x and so preserves this. For the last statement, if $\Im w > \sqrt{|D|}/2$ then the identity coset contributes nothing, and only the finitely many cosets $\gamma \in \Gamma_\infty\backslash\Gamma$ with $\Im\gamma w \le \sqrt{|D|}/2$ remain; on the rest of a fundamental domain the sum is dominated by $tE(w, 2)$, which is bounded there. So $P[\Phi_t^\flat]$ is bounded, hence square-integrable on the finite-volume quotient. □

[6] Verified for $D \in \{-11, -8, -7, -4, -3, 5, 8, 12\}$ and all odd $d \le 200$ coprime to $2D$, and $k \le 3$.

Remark 16 (the continuous spectrum). Property (iii) also disposes of the Eisenstein contribution. Unfolding gives $\langle P[\Phi_t^\flat], E(\cdot, \frac{1}{2}+ir)\rangle = \int_{\mathbb{H}} \Phi_t^\flat E(\cdot, \frac{1}{2}+ir)\, d\mu$, and the two terms y^s and $\varphi(s)y^{1-s}$ of the Fourier expansion of E do not depend on x, so by (iii) they contribute nothing. Only the non-constant modes of E remain, and against those the computation of Step 4 is the same one with it_j replaced by ir. The continuous spectrum therefore contributes $\sqrt{Y}\int c(r)Y^{ir}\,dr$ to the Riesz mean, which is $o(\sqrt{Y})$ by the Riemann–Lebesgue lemma as soon as $c \in L^1$. This is why the discrete lines are what one sees.

Proposition 17 (which Riesz order)**.** *Normalise u_j in $L^2(\Gamma\backslash\mathbb{H})$ and write $a_j(n) = \rho_j(1)\lambda_j(n)$ with λ_j the Hecke eigenvalues. Then in* (21)

$$|A_j(D)| \;\asymp_D\; t_j^{-1/2+o(1)}, \tag{22}$$

and the Riesz mean of order m multiplies the j-th term by a factor $\asymp t_j^{-m-1}$. Consequently the spectral sum of Theorem 8 converges absolutely as soon as

$m \;\geq\; 1$ *(using $\|u_j\|_\infty \ll t_j^{5/12+\varepsilon}$ on compact sets, Iwaniec–Sarnak),* *or* $m \;\geq\; 2$ *(using only $\|u_j\|_\infty <$*

In particular the Cesàro mean, $m = 1$, suffices, which is the mean at which the numerics of §6 were computed.

Proof. For (22): the k-sum $\sum_k a_j(2k)k^{-3/2} = \rho_j(1)\sum_k \lambda_j(2k)k^{-3/2}$ converges absolutely and is $\rho_j(1)t_j^{o(1)}$, since $\lambda_j(n) \ll n^{7/64+\varepsilon}$ and $\frac{3}{2} - \frac{7}{64} > 1$. The Γ-factors of (21) have modulus

$$|\Gamma(-it_j)|\,\left|\Gamma(\tfrac{1}{2}+it_j)\right|\,\left|\sin\tfrac{\pi(\frac{1}{2}+it_j)}{2}\right| \asymp \sqrt{\frac{2\pi}{t_j}}e^{-\pi t_j/2}\cdot\sqrt{2\pi}e^{-\pi t_j/2}\cdot\frac{e^{\pi t_j/2}}{2} \asymp \frac{1}{\sqrt{t_j}}\,e^{-\pi t_j/2},$$

while $|\rho_j(1)|^2 \asymp \cosh(\pi t_j)/L(1,\mathrm{sym}^2 u_j)$ gives $|\rho_j(1)| \asymp e^{\pi t_j/2}t_j^{o(1)}$. The two exponentials cancel and (22) follows. For the Riesz factor, $\frac{1}{m!}\int_0^Y (Y-t)^m t^{-1/2-it_j}\,dt = \frac{Y^{m+1/2-it_j}}{m!}B(m+1,\frac{1}{2}-it_j)$ and $B(m+1,\sigma) = \Gamma(m+1)\Gamma(\sigma)/\Gamma(m+1+\sigma) \asymp |\sigma|^{-m-1}$. Finally $|\mathrm{Per}_D(u_j)| \leq h(D)\|u_j\|_\infty$, so the j-th amplitude is $\ll t_j^{5/12+\varepsilon}\cdot t_j^{-1/2+o(1)}\cdot t_j^{-m-1}$, and by the Weyl law $\#\{j : t_j \leq T\} \sim T^2/12$ the sum over j is dominated by $\int^\infty t^{5/12-1/2-m-1}\,t\,dt = \int^\infty t^{-m-1/12}\,dt$, which converges for $m > \frac{11}{12}$. With the trivial sup-norm bound the exponent is $\int^\infty t^{-m}dt$ instead. $\square$

The expansion (20)**, uniformly in the order.** One point remains: (20) is an asymptotic as $x \to 0$ *for fixed order*, whereas we apply it at $x = \sqrt{|D|}\,w/t$ with t_j ranging over the whole spectrum. What is needed is a version uniform in t_j. It is elementary, and follows from the convergent series for K_ν rather than from any uniform-asymptotic machinery.

Lemma 18 (uniform small-argument expansion)**.** *For $\tau \geq 1$ and $x > 0$,*

$$K_{i\tau}(x) = \tfrac{1}{2}\Big[\Gamma(-i\tau)\big(\tfrac{x}{2}\big)^{i\tau} + \Gamma(i\tau)\big(\tfrac{x}{2}\big)^{-i\tau}\Big] + E(x,\tau), \qquad |E(x,\tau)| \;\leq\; |\Gamma(i\tau)|\,\big(e^{x^2/(4\tau)} - 1\big).$$

In particular $|E(x,\tau)| \leq |\Gamma(i\tau)|\,x^2/(2\tau)$ whenever $x^2 \leq 2\tau$.

Proof. From $K_\nu = \frac{\pi}{2\sin\nu\pi}(I_{-\nu} - I_\nu)$, the series $I_\nu(x) = \sum_{n\geq 0}(x/2)^{2n+\nu}/(n!\,\Gamma(n+\nu+1))$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$,

$$K_{i\tau}(x) = \tfrac{1}{2}\Gamma(i\tau)\Gamma(1-i\tau)\sum_{n\geq 0}\frac{(x/2)^{2n}}{n!}\Big[\frac{(x/2)^{-i\tau}}{\Gamma(n+1-i\tau)} - \frac{(x/2)^{i\tau}}{\Gamma(n+1+i\tau)}\Big],$$

convergent for every x. The term $n = 0$ is the stated main term, using $\Gamma(1-i\tau) = -i\tau\,\Gamma(-i\tau)$. For $n \geq 1$,

$$\Big|\frac{\Gamma(1\mp i\tau)}{\Gamma(n+1\mp i\tau)}\Big| = \frac{1}{|(1\mp i\tau)(2\mp i\tau)\cdots(n\mp i\tau)|} \leq \tau^{-n},$$

since each factor has modulus at least τ. As $|\Gamma(i\tau)| = |\Gamma(-i\tau)|$, the tail is at most $|\Gamma(i\tau)| \sum_{n\ge1}(x^2/4\tau)^n/n! = |\Gamma(i\tau)|(e^{x^2/(4\tau)} - 1)$. □

[7]

Lemma 18 is exactly what (21) needs. In I_k the argument is $x = cw$ with $c = \alpha_k/\beta_k = \sqrt{|D|}/t$, the same for every k; splitting the w-integral at $W = \sqrt{t_j}/c$ and applying the lemma below W gives an error

$$\ll |\Gamma(it_j)|\,\frac{c^2}{t_j}\int_0^W w^{3/2}\,dw \;\asymp\; |\Gamma(it_j)|\,c^{-1/2}\,t_j^{-3/4},$$

against a main term of size $\asymp |\Gamma(it_j)|c^{-1/2}$: a saving of $t_j^{-3/4}$, uniform in j, which Proposition 17 has ample room to absorb. Note also that the stationary point of the combined phase, where $t_j\,d(\log x)/dx$ meets $1/c$, sits at $x = \sqrt{|D|}$, a fixed point independent of t_j and well inside the range where the lemma applies.

5.2 The remaining estimates, and the proof

Two estimates finish the argument: an exact treatment of the integral I_k, and a bound for the tail removed by the truncation. Both are elementary. Throughout, the truncation height of Lemma 15 is taken to be $T = t$ rather than $\sqrt{|D|}/2$; this is permitted, since any height above $\sqrt{|D|}/2$ leaves the orbit sum unchanged, and it costs only $\|P[\Phi_t^\flat]\|_2 \ll \sqrt{t}$.

Lemma 19 (the integral, exactly)**.** *Let $\alpha, \beta > 0$, $\tau \ge 1$, and $I(\alpha,\beta,\tau) = \int_0^\infty y^{-1/2}\sin(\beta y)K_{i\tau}(\alpha y)\,dy$. Then for any $\varepsilon \in (0, \frac12)$,*

$$I = \beta^{-1/2}\,\Re\Big[\Gamma(-i\tau)\,\Gamma\big(\tfrac12 + i\tau\big)\sin\tfrac{\pi(\frac12+i\tau)}{2}\Big(\frac{\alpha}{2\beta}\Big)^{i\tau}\Big] \;+\; O_\varepsilon\Big(\beta^{-1/2}\,e^{-\pi\tau/2}\,\tau^A\Big(\frac{\alpha}{\beta}\Big)^{2-\varepsilon}\Big)$$

for an absolute constant A, uniformly in α, β, τ.

Proof. The Mellin transforms $\int_0^\infty y^{s-1}\sin(\beta y)\,dy = \Gamma(s)\sin(\pi s/2)\beta^{-s}$, valid for $0 < \Re s < 1$, and $\int_0^\infty y^{s-1}K_{i\tau}(\alpha y)\,dy = 2^{s-2}\alpha^{-s}\Gamma(\frac{s+i\tau}{2})\Gamma(\frac{s-i\tau}{2})$, valid for $\Re s > 0$, give by Parseval's formula for the Mellin transform, for any $c \in (0, \frac12)$,

$$I = \frac{1}{2\pi i}\int_{(c)} \Gamma(s)\sin\frac{\pi s}{2}\,\beta^{-s}\,2^{-\frac32 - s}\,\alpha^{s-\frac12}\,\Gamma\Big(\frac{\frac12 - s + i\tau}{2}\Big)\Gamma\Big(\frac{\frac12 - s - i\tau}{2}\Big)ds. \tag{23}$$

In $\Re s > 0$ the integrand has poles only where the last two Γ-factors do, namely at $s = \frac12 \pm i\tau + 2n$, $n \ge 0$; $\Gamma(s)$ contributes none there and $\sin(\pi s/2)$ contributes none at all. Move the contour to $\Re s = \frac52 - \varepsilon$, crossing exactly the two poles with $n = 0$. Their residues are the displayed main term, and on the new line $|\beta^{-s}\alpha^{s-1/2}| = \beta^{-1/2}(\alpha/\beta)^{2-\varepsilon}$, which is the stated saving. That the shifted integral converges, and that its size is $e^{-\pi\tau/2}\tau^A$ times the above, follows from Stirling's formula: on $\Re s = \frac52 - \varepsilon$ the two Γ-factors have argument of real part $-1 + \frac{\varepsilon}{2}$ and imaginary parts $\frac12(\pm\tau - \Im s)$, so their product decays like $e^{-\frac{\pi}{4}(|\tau - \Im s| + |\tau + \Im s|)} \le e^{-\pi\tau/2}$ times a power of $1 + |s| + \tau$, while $\Gamma(s)\sin(\pi s/2)$ grows at most like a power times $e^{-\pi|\Im s|/2}\cdot e^{\pi|\Im s|/2}$; the resulting integral in $\Im s$ converges absolutely. [8] □

[7]The bound was checked numerically at 30 pairs with $\tau \in \{2, 5, 13.78, 40, 100, 300\}$ and $x \in \{0.5, 1, 2, 4, 8\}$; it holds in every case, with ratio between 0.0002 and 0.995.

[8]The main term was checked against numerical quadrature: for $\tau = 13.7798$ and $\alpha/\beta = 0.1, 0.04, 0.02, 0.01$ the relative error is $0.0715, 0.0099, 0.0018, 0.00035$, i.e. $\asymp (\alpha/\beta)^2$ with a bounded constant, confirming both the main term and the exponent.

Lemma 20 (the truncated tail). *For $\alpha, \beta > 0$, $\tau \geq 1$ and $\alpha T \geq 1$,*

$$\Big|\int_T^\infty y^{-1/2} \sin(\beta y) K_{i\tau}(\alpha y)\, dy\Big| \ll \alpha^{-1/2}(\alpha T)^{-1/2} e^{-\alpha T},$$

uniformly in β and τ.

Proof. From $K_\nu(x) = \int_0^\infty e^{-x\cosh u}\cosh(\nu u)\, du$ one gets, for $\nu = i\tau$, $|K_{i\tau}(x)| \leq \int_0^\infty e^{-x\cosh u} du = K_0(x)$. [9] Hence the integral is at most $\int_T^\infty y^{-1/2} K_0(\alpha y)\, dy$, and substituting $x = \alpha y$ and using $K_0(x) \ll x^{-1/2}e^{-x}$ for $x \geq 1$ gives the claim. □

Theorem 21. *Let $D < 0$ be a discriminant, $u = 1$, and let $\mathcal{G}^{(m)}(Y)$ denote the Riesz mean of order m of the model object (24). For $m \geq 2$ there are coefficients $\alpha_j(D)$, supported on the even Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$ and proportional to the Katok–Sarnak periods $\mathrm{Per}_D(u_j)$, such that*

$$\mathcal{G}^{(m)}(Y) = Y^m\sqrt{Y}\sum_j \big(\alpha_j(D)\, Y^{it_j} + \overline{\alpha_j(D)}\, Y^{-it_j}\big) \; + \; \mathcal{E}(Y) \; + \; O\big(Y^{m+\frac{1}{2}-\delta}\big)$$

for some $\delta > 0$, the sum converging absolutely, where $\mathcal{E}$ is the contribution of the continuous spectrum and satisfies $\mathcal{E}(Y) = o(Y^{m+1/2})$.

Proof. Apply the Riesz mean of order m to (16) and truncate the seed at height $T = t$ (Lemma 15), which changes nothing. By Lemma 13 the automorphised seed converges absolutely and lies in $L^2(\Gamma\backslash\mathbb{H})$, so it has a spectral decomposition; by Proposition 17 the Riesz factor $\asymp t_j^{-m-1}$ makes the discrete part converge absolutely for $m \geq 2$ even with only the trivial sup-norm bound, so the expansion converges uniformly and may be evaluated at the Heegner points and summed over the orbit, which replaces u_j by $\mathrm{Per}_D(u_j)$ (Step 3). The constant term vanishes because $\int_0^1 \Phi_t^\flat\, dx = 0$ (Lemma 15(iii)), and the same identity confines the Eisenstein contribution to the non-constant modes, whence $\mathcal{E}(Y) = o(Y^{m+1/2})$ by Remark 16.

For the coefficients, (19) expresses $\langle \Phi_t^\flat, u_j\rangle$ as $\sum_k (a_j(2k)/\pi k)$ times $I(\alpha_k, \beta_k, t_j)$ minus the truncated tail, with $\alpha_k = 4\pi k$ and $\beta_k = 4\pi kt/\sqrt{|D|}$. The interchange of the k-sum with the y-integral is justified by $\int_0^\infty y^{-1/2}K_0(\alpha_k y)\, dy \ll k^{-1/2}$ together with $\sum_k |a_j(2k)|k^{-3/2} < \infty$, which holds because $\lambda_j(n) \ll n^{7/64+\varepsilon}$. Lemma 19 evaluates each $I(\alpha_k, \beta_k, t_j)$ with a relative error $O(\tau^A(\alpha_k/\beta_k)^{2-\varepsilon})$; crucially $\alpha_k/\beta_k = \sqrt{|D|}/t$ does not depend on k, so the relative error is uniform in k and survives the k-sum unchanged. Lemma 20 bounds the tail by $\ll e^{-4\pi kt}$, which after summation over k and, using the Riesz factor, over j, is $O(e^{-2\pi t})$. Collecting the main terms gives $\langle \Phi_t^\flat, u_j\rangle = t^{-1/2}(A_j(D)t^{-it_j} + \overline{A_j(D)}t^{it_j}) + O(t^{-5/2+\varepsilon}t_j^A)$ with $|A_j(D)| \asymp_D t_j^{-1/2+o(1)}$ by (22); the factor $(\alpha_k/2\beta_k)^{it_j} = (\sqrt{|D|}/2t)^{it_j}$ supplies the dependence on the discriminant. Summing $t^{-1/2\mp it_j}$ against the Riesz weight turns it into $Y^{m+1/2\mp it_j}$ times $B(m+1, \frac{1}{2} \mp it_j) \asymp t_j^{-m-1}$, and the error terms sum by Proposition 17 with room to spare, since they carry two further powers of t. □

Remark 22 (what a referee should check). The proof uses, as standard inputs: Gauss's parametrisation of the root pairs by binary quadratic forms (§5, Step 1), the spectral decomposition of $L^2(\Gamma\backslash\mathbb{H})$, the Weyl law, the sup-norm bound for Maass forms, the bound $\lambda_j(n) \ll n^{7/64+\varepsilon}$, and Parseval's formula for the Mellin transform. Everything else is proved above. The two places we would look first for an error are the uniformity in τ of the shifted contour in Lemma 19, where we have given the Stirling estimate in outline rather than in full, and the passage from the spectral expansion of $P[\Phi_t^\flat]$ in L^2 to its pointwise evaluation at the Heegner points, which we justify by the absolute convergence supplied by the Riesz factor and which is the reason the theorem is stated for $m \geq 2$ rather than $m \geq 1$. This paper has twice contained an argument that appealed to a classical theorem about a different object (Remarks 7 and 12); we would rather name the weak points than let a reader hunt for them.

[9] Checked at 25 pairs, $\tau \leq 120$, $x \leq 100$: the inequality holds in every case.

6 Numerical evidence

All computations are exact rational or high-precision arithmetic; the scripts are in the repository (`research/paper-IV/scripts/`). For each f and u we compute $\mathcal{P}_u^{(1)}(Y)$ for all $Y \le 10^7$ (a sieve of $Q_u(h)$ for $h \le Y$, with $\mathbb{E}F_u$ evaluated to 22 significant digits, since an error δ in $\mathbb{E}F_u$ enters as $\delta Y^2/2$), sample it on a logarithmic grid of 4096 points in $\log Y \in [\log 10^3, \log Y_{\max}]$, remove the mean and the linear trend in $\log Y$, and regress the result on

$$\{\cos(t_j \log Y),\ \sin(t_j \log Y)\}_{j=1}^{6}$$

for the six smallest even spectral parameters, for the six smallest odd ones, and for 300 random sets of six frequencies drawn uniformly from $[12, 25]$. Reported is the fraction of variance explained, R^2, and where it falls among the random sets.

Table 1: Fit of $\mathcal{P}_u^{(1)}(Y)/\sqrt{Y}$ to the even and to the odd spectral parameters of $\mathrm{SL}_2(\mathbb{Z})$, against 300 random frequency sets of the same size ($Y \le 10^7$; $u = 1$ throughout except where shown).

f	D	R^2 even	percentile	R^2 odd	percentile	verdict
t^2-2	8	0.217	100	0.036	34	even
t^2+1	-4	0.152	100	0.028	20	even
t^2+t+1	-3	0.139	100	0.037	35	even
t^2+t+2	-7	0.089	100	0.051	76	even
t^2-3	12	0.066	99	0.037	66	even
t^2+t-1	5	0.062	92	0.036	32	even
t^2+t+5	-19	0.047	92	0.044	88	inconclusive
t^2+2	-8	0.031	47	0.054	86	none
t^2+t+3	-11	0.009	44	0.023	100	none

Table 1 is the evidence for Proposition 9. Six of the nine polynomials place the even set at the 92nd percentile or above, four of them above every one of the 300 random sets; the odd set never rises above the 88th percentile except once, at the 100th for t^2+t+3, where the even set is at the 44th and the fit is at the level of the noise. For the dominant line $t_1 = 13.7798$ the fitted amplitude and phase are stable when the range of $\log Y$ is cut in half: for t^2-2, amplitudes 0.019 and 0.018 with phases 0.89 and 0.61; for t^2+t+1, 0.021 and 0.017 with phases 1.01 and 1.03; for t^2+1 at $u=2$, 0.042 and 0.042 with phases -0.30 and 0.07. A spurious fit does not reproduce its phase on an independent range.

The periods. Using the Fourier coefficients of the first even form (LMFDB label `1.0.1.3.1`, $t_1 = 13.7797513$) we evaluated u_1 at the Heegner points of the nine imaginary discriminants of class number one (`scripts/maass-period.py`).

Table 2: The Katok–Sarnak period of the first even Maass form at the Heegner point of discriminant $D < 0$, class number one, relative to $D = -4$; and the fitted amplitude of the line t_1, relative to $D = -4$.

D	-3	-4	-7	-8	-11	-19	-43	-67	-163
$\lvert u_1(z_D)\rvert/\lvert u_1(i)\rvert$	0.649	1	0.542	0.318	0.302	0.444	$6.6\cdot10^{-3}$	$1.0\cdot10^{-4}$	$2.2\cdot10^{-10}$
fitted amplitude ratio	0.483	1	0.111	0.281	0.461	1.358	—	—	—
signal present	yes	yes	yes	no	no	marginal	no	no	no

Table 2 confirms the mechanism at the level of orders of magnitude and no further. The threshold (13) sits at $|D| \approx 19.2$: below it every period lies within a factor 3 of every other, and every one of the six discriminants with $|D| \le 19$ has an amplitude within a factor 12 of the others; above it the periods fall by two, four and ten orders of magnitude at $D = -43, -67, -163$, and no signal is seen at any of them. In particular the absence at $D = -163$, the polynomial t^2+t+41, is explained. What the table does *not* confirm is the amplitude law itself: the two rows disagree by as much as a factor 5 at $D = -7$. Two unmodelled effects are of that size, the local factors, since $\mathbb{E}F_u$ ranges over $[0.35, 1.60]$ across these polynomials and enters multiplicatively, and the uncertainty of the fitted amplitudes themselves, which for $D = -7$ is a factor 2 between the two halves of the range. We record the amplitude law as a prediction to be tested, not as a confirmed fact.

The weight, and a model piece. The absences at $D = -8$ and $D = -11$, whose periods are of ordinary size, are not explained by anything above, and they are not isolated: among the fourteen polynomials we tested, the six showing no signal are exactly those for which the prime 3 splits in $\mathbb{Q}(\sqrt{D})$. That is a statement about the weight, not about the roots. Indeed $\lambda(3) = 3/(3-4) = -3$ is the only negative value taken by λ and the largest in modulus, and $|\lambda(3)\rho(3)/3| = 2 > 1$, so at $\Re s = \frac{1}{2}$ the local factor at 3 of the Euler product for W_k has modulus up to $3 \cdot 2 \cdot 3^{-1/2} = 3.46$: it is the one local factor that can dominate the spectral term. Removing the weight confirms this. Define the *model piece* by replacing λ by 1 in (1), so that F becomes the restricted divisor function $\sum_{d|n,\ d \text{ adm}} 1/d$ and $\mathcal{P}^{(1)}$ is a Riesz mean of σ_{-1} along Q_u. Repeating the regression:

Table 3: Removing the arithmetic weight, at $Y \leq 10^7$, $u = 1$. Fit of $\mathcal{P}^{(1)}/\sqrt{Y}$ to the six even spectral parameters, for the weighted piece of part III, for the same sum with $\lambda \equiv 1$, and for the sum over all divisors rather than the squarefree ones. The odd sets, omitted, run from the 4.7th to the 28.7th percentile throughout.

	weighted, squarefree		$\lambda \equiv 1$, squarefree		$\lambda \equiv 1$, all divisors	
D	R^2 even	pct	R^2 even	pct	R^2 even	pct
8	0.217	100	0.319	100	0.420	100
-4	0.152	100	0.237	100	0.361	100
-3	0.139	100	0.139	100	0.208	100
-8	0.031	47	0.109	100	0.162	100
-11	0.009	44	0.078	100	0.167	100
17	0.078	99.3	0.098	99.3	0.116	100

Every entry of the last four columns is at the 99.3rd percentile or above, every odd set falls below the 29th, and the two discriminants that showed nothing under the weight show a clean signal without it. So the apparent rule at $p = 3$ is an artefact of λ; nothing arithmetic is hidden there. Note also that the fit improves monotonically along the three columns, for every one of the six discriminants: the more of the arithmetic weight one removes, the more of the variance the even spectrum explains. That is itself evidence that the spectral term is real and that the weight is what obscures it.

The last column is the cleanest object of all. Dropping both the weight and the squarefree condition leaves

$$\mathcal{G}(Y) = \sum_{h \leq Y} (Y - h)\big(\sigma^*_{-1}(h^2 - D) - \mathbb{E}\sigma^*_{-1}\big), \qquad \sigma^*_{-1}(n) = \sum_{\substack{d|n \\ (d,2D)=1}} \frac{1}{d}, \tag{24}$$

a Riesz mean of the divisor-type sums of Hooley and Gafurov. Lemmas 1 and 2 never used λ or squarefreeness, so they apply verbatim, and the Dirichlet series attached to (24) is

$$Z_k(s) = \sum_{(d,2D)=1} \mathcal{W}_k(D; d)\, d^{-s},$$

a Salié zeta function carrying a coprimality condition and nothing else. This is the object of §5. The piece the off-diagonal of part III actually requires is (24) perturbed twice: by the squarefree condition, through Möbius over ℓ^2, and by $\lambda = 1 * \kappa$ with $\kappa(p) = 4/(p-4) = O(1/p)$.

The phases: a test with no free parameter. The sharpest check available to us is not the size of the oscillation but its phase. In (16) the length enters the test function only through the combination $2t/\sqrt{|D|}$, so the spectral term carries the factor $(2/\sqrt{|D|})^{it_j}$; and by Step 3 the amplitude is proportional to $\mathrm{Per}_D(u_j)$, a real number whose *sign* therefore shifts the phase by π. Writing the fitted oscillation as $A_D \cos(t_1 \log Y - \varphi_D)$, the mechanism predicts, once one discriminant is fixed as a reference and with nothing else at our disposal,

$$\varphi_D - \varphi_{D_0} = \frac{t_1}{2} \log \frac{|D|}{|D_0|} - \pi \cdot \left[\operatorname{sign} \mathrm{Per}_D \neq \operatorname{sign} \mathrm{Per}_{D_0}\right] \pmod{2\pi}. \tag{25}$$

The signs of the periods are computed from the Fourier coefficients of the first even form, independently of the arithmetic data. Taking $D_0 = -4$ (`scripts/phase-test.py`):

Table 4: The phase of the $t_1 = 13.7798$ oscillation of the model object, predicted by (25) with no free parameter, against the phase fitted to the data ($Y \le 10^7$, $u = 1$). The reference $D_0 = -4$ is fitted, the other five are predictions.

D	sign Per_D	predicted φ_D	observed φ_D	difference
-3	$+$	$+1.158$	$+1.050$	$+0.108$
-4	$+$	$+3.140$	$+3.140$	(reference)
-7	$+$	$+0.712$	$+0.500$	$+0.212$
-8	$-$	-1.509	-1.430	-0.079
-11	$-$	$+0.685$	$+0.530$	$+0.155$
-19	$-$	-1.833	-1.870	$+0.037$

Every one of the five predictions is correct to within 0.212 radians, that is to 3.4% of a full period. Five independent phases falling that close to prediction by chance has probability about $1.4 \cdot 10^{-6}$. The test uses both halves of the mechanism at once: the factor $(2/\sqrt{|D|})^{it_1}$, which is what makes the predicted phases differ from one another at all, and the sign of the Katok–Sarnak period, which is what puts $D = -8, -11, -19$ on the opposite side. We regard this, rather than any of the variance fits, as the evidence that the mechanism of §5 is the right one.

For the amplitudes the agreement is weaker but not absent. Dividing the fitted amplitude by $|\mathrm{Per}_D(u_1)|/|\Gamma_{z_D}|$ gives, in units of 10^7, the values 2.68, 2.22, 0.39, 1.90, 2.71, 2.71 for $D = -3, -4, -7, -8, -11, -19$: five of the six agree to within 40% with no further factor, and the outlier is $D = -7$, which is also the case whose fitted amplitude is least stable, moving by a factor 1.7 between the two halves of the range.

6.1 The same test on Hooley's sum

Everything above concerns a divisor sum smoothed by a Riesz mean. The identity (17) predicts the same law for a barer object, namely the partial sums of the Weyl sums themselves,

$$T_k(X) = \sum_{\substack{d \le X \\ (d,2D)=1}} \; \sum_{b \bmod d,\; b^2 \equiv D} \mathrm{e}\Big(\frac{kb}{d}\Big),$$

the sum Hooley bounded by $O(X^{3/4+\varepsilon})$ in [5]. Since the poles of its Dirichlet series are those of (17), at $s = \frac{1}{2} \pm it_j$, the prediction is $T_k(X) = \sqrt{X} \sum_j c_j X^{it_j} + \cdots$, with no smoothing anywhere. We computed $T_1(X)$ for all $X \le 10^6$ (`scripts/weyl-partial.ts`; roots of $b^2 \equiv D$ by Tonelli–Shanks, Hensel and the Chinese remainder theorem).

$T_1(X)/\sqrt{X}$ is indeed bounded, staying below 0.56 in absolute value with standard deviation 0.14 for $D = -4$: square-root cancellation, well beyond the proved exponent $\frac{3}{4}$. Its spectrum is the even one, for every discriminant tested, at the 96th percentile or above, while the frequencies $2t_j$ are rejected at the 12.7th. And the phase test of (25) applies verbatim, with $(2/\sqrt{|D|})^{it_1}$ entering through the factor $(\sqrt{|D|}/2)^s$ of (17):

Table 5: The same parameter-free phase prediction (25), applied to the partial sums of Hooley's Weyl sums $T_1(X)$, $X \le 10^6$. Reference $D_0 = -4$; the other five are predictions.

D	R^2 even	percentile	predicted φ_D	observed φ_D	difference
-3	0.173	100	-2.992	-2.980	-0.012
-4	0.412	100	-1.010	-1.010	(reference)
-7	0.159	100	$+2.846$	$+3.110$	-0.264
-8	0.252	98.3	$+0.624$	$+0.520$	$+0.104$
-11	0.170	100	$+2.818$	$+2.810$	$+0.008$
-19	0.206	96.0	$+0.301$	$+0.310$	-0.009

Three of the five predictions are correct to within 0.012 radians, two parts in a thousand of a full period, and the median discrepancy over the five is 0.012. We know of no way to obtain agreement of this kind other than by the mechanism of §5 being correct.

The limits of the present computations, and what is planned. Every number reported above comes from a sieve of $Q_u(h)$ for $h \le 10^7$, which runs in seconds to minutes on a laptop; the scripts are in the repository and each states its cost in its header. Three of the questions raised here are limited by that range rather than by anything conceptual, and are settled by longer runs of exactly the same code:

1. the amplitude law of Table 2, whose comparison is currently blunted by an uncertainty of up to a factor 2 in the fitted amplitudes; the uncertainty falls like $Y^{-1/2}$ in the length of the $\log Y$ range, so $Y = 10^9$ would roughly halve it and $Y = 10^{12}$ would settle the discrepancy at $D = -7$;

2. the resolution in the frequency, which is $2\pi/\log(Y_{\max}/Y_{\min}) \approx 0.68$ at present and therefore cannot separate the lines 19.42 and 19.48, the first even and the first odd parameter of that size; this is the single measurement that would test the parity rule against an adjacent pair rather than against a whole set;

3. the selection rule at $p = 3$, where the sample is fourteen polynomials and should be a few hundred.

None of the three needs a new method, only more arithmetic, and we intend to carry the computation out on appropriate hardware and report it separately. We state them here so that a reader can see exactly which of our assertions rest on 10^7 and which do not: the parity rule of Table 1 and the threshold of Table 2 are stable across the halves of the present range and do not, the amplitude law does.

7 Uniformity in u, and what the Cesàro conjecture would need

Theorem 21 is for one piece, $u = 1$. Part III reduces the Cesàro form of Conjecture 1 of part I to the windows of *all* the pieces with $u \le H^{2/3+\varepsilon}$, so what is wanted is Theorem 21 uniformly in

u, with an error that survives summation against the weight $w(u)$ over that range. This section records what that requires, and why the proof above does not supply it.

For $u > 1$ the roots are those of $u^2x^2 \equiv D$, and by Lemma 5 the dilation moves the frequency from k to $k\bar{u}$, a frequency depending on the modulus. Geometrically the change is the same one: by Lemma 5 and (14) the pairs (d, x) with $u^2x^2 \equiv D$ (d) are again Heegner points, now of discriminant $4u^2D$, so the orbit lies at level $4u^2$ rather than level 1. Everything in §5.2 then goes through with Γ replaced by $\Gamma_0(4u^2)$, except for two things, and both are the same thing in different clothing:

1. the sup-norm and Weyl-law inputs of Proposition 17 acquire a dependence on the level, and
2. the periods $\mathrm{Per}_{4u^2D}(u_j)$ are taken over $h(4u^2D) \asymp u^{1+o(1)}$ classes rather than $h(D)$.

Summing $\lambda\omega(u)$ times an error $(H/u)^{1/2}u^A$ over $u \le H^{2/3}$ is $O(H^{1-\delta})$ precisely when $A < \frac{1}{4}$. So the question is whether the spectral estimates can be made uniform in the level with a loss below $u^{1/4}$. The spectral large sieve of Deshouillers and Iwaniec is uniform in the level in the shape one wants, but the sup-norm bound for individual forms is not, and the periods at level $4u^2$ are exactly the objects that Katok–Sarnak controls only for fixed discriminant. This is the same wall that part III meets from the other side, in the guise of the exponent condition $\theta + 6B < 1$: there, no Weyl-sum bound is uniform enough in the frequency; here, no spectral bound is uniform enough in the level. We regard the two as the same open problem seen from two directions, and we do not know how to cross it.

What the present paper does say about it is that the object on the far side is now identified. The windows of part III are not an amorphous error term: piece u carries the spectrum of level $4u^2$, with amplitudes the periods at discriminant $4u^2D$, and the Cesàro conjecture is the statement that these spectra, summed over u with the weight $w(u)/u$, exhibit enough cancellation. That is a sharper question than the one part III could pose, and it is the one we would attack next.

AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model was used to draft mathematical arguments, to write and run the software behind every number in the paper, to search the literature, and to draft the text; all of this was done in dialogue with the author, who verified the code and the numerical results, reviewed the proofs, and takes full responsibility for the content. The model is not an author. The complete record of the work is in the public repository `https://github.com/gewure/ulamnd` (directory `research/`).

References

[1] V. A. Bykovskiĭ. Spectral expansions of certain automorphic functions and their number-theoretic applications. *Zap. Nauchn. Sem. LOMI*, 134:15–33, 1984. English transl.: J. Soviet Math. 36 (1987), 8–21.

[2] W. Duke, J. B. Friedlander, and H. Iwaniec. Equidistribution of roots of a quadratic congruence to prime moduli. *Annals of Mathematics*, 141:423–441, 1995.

[3] W. Duke, J. B. Friedlander, and H. Iwaniec. Weyl sums for quadratic roots. *Int. Math. Res. Not. IMRN*, (11):2493–2549, 2012.

[4] Dorian Goldfeld and Peter Sarnak. Sums of Kloosterman sums. *Inventiones Mathematicae*, 71:243–250, 1983.

[5] C. Hooley. On the number of divisors of quadratic polynomials. *Acta Mathematica*, 110:97–114, 1963.

[6] Henryk Iwaniec. *Topics in Classical Automorphic Forms*, volume 17 of *Graduate Studies in Mathematics*. American Mathematical Society, 1997.

[7] Svetlana Katok and Peter Sarnak. Heegner points, cycles and Maass forms. *Israel Journal of Mathematics*, 84:193–227, 1993.

[8] Matthew Welsh. Parameterizing roots of polynomial congruences. *Algebra & Number Theory*, 16:881–918, 2022.